Project Title: FitTrack Pro Fitness Management System
Course: COCS 440, Database Management Systems
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Purpose

The goal of this project is to design and implement a relational database for a fitness management website. The database supports user accounts, workout programs, exercises, trainers, group classes, nutrition logs, and progress tracking. The system focuses on clean structure, data integrity, and real usage in a web app.

Database Design

The database uses 13 tables in Third Normal Form.

Key tables are users, user_profiles, trainers, workout_programs, exercises, user_workouts, workout_exercises, goals, nutrition_logs, progress_tracking, class_schedules, class_enrollments, and program_exercises.

Each table has a primary key. Foreign keys link related records and protect referential integrity.

Many to many links use junction tables. Examples are program_exercises for programs and exercises, and class_enrollments for users and classes.

Main Features

User management

- Separate tables for login data and fitness profile data
- One to one link between users and user_profiles
- Goals, workouts, nutrition logs, and progress entries linked to each user

Workout programs and exercises

- Trainers create workout_programs
- Each program links to many exercises through program_exercises
- Difficulty level and duration stored at program level
- Exercise details stored once in exercises and reused in programs and workout logs

Workout tracking

- Each workout session stored in user_workouts
- Each session links to detailed sets and reps in workout_exercises

- Design supports tracking weight used, intensity, and notes for each exercise

Classes and enrollments

- Group classes stored in class_schedules with trainer, date, time, location, and capacity
- Enrollments stored in class_enrollments with status, attendance, and rating
- Structure supports real time calculation of open spots and attendance history

Normalization and integrity

- All tables follow 3NF, no repeating groups, and no partial or transitive dependencies
- Foreign keys prevent orphan records for users, trainers, classes, and programs
- Unique constraint on user email and on each user and class pair in class_enrollments
- Boolean flags such as is_active and attended support clear business rules

Implementation Summary
The backend uses MySQL with PHP for API endpoints.
Core endpoints return JSON for programs, trainers, exercises, and classes.
The frontend uses JavaScript to fetch data and update the interface without page reloads.
Validation runs on the client and the server to protect the database and improve user experience.

What this system shows

- A complete ER diagram that matches the MySQL schema
- Correct use of primary keys, foreign keys, and junction tables
- Support for realistic fitness workflows
- A database design ready for extension with features such as progress charts or payments